

LESSON 4

LINEAR CORRELATION AND REGRESSION

Overview:

There are two major areas of statistics: Descriptive Statistics and Inferential Statistics. The former includes Measures of Central Tendency, Dispersion, and Relative Position which were discussed in lessons 1 to 3. This lesson will involve correlation and regression analysis that are used in inferential statistics. We shall be studying two variables at one time or simultaneously. We shall see how one set of data is related to another.

Learning Outcomes:

At the end of this lesson, the students can:

1. draw a scatter plot for a set of variables;
2. identify the relationship of the given variables;
3. give examples of variables with different relationships;
4. compute the correlation coefficient;
5. interpret the obtained correlation coefficient; and
6. determine the regression equation.

Materials Needed:

- learning module
- ruler
- scientific calculator.

Duration: 3 hours

Learning Contents:

4.1 CORRELATION

Correlation is an analysis that measures the *strength of association* and the *direction of relationship* between two variables. It is used to test relationships between quantitative variables. In other words, it is a measure of how two things are related to each other. By studying the relationship of these variables, we can make predictions on their possible future behavior.

It is a statistical method used to determine whether a relationship between two variables exists. There are two variables involve in this statistical method:

Independent Variable, a predictor variable; and
Dependent Variable, a response or outcome variable.

4.2 TYPES OF RELATIONSHIP

- 1. **Positive Relationship** exists when both variables increase or decrease at the same time.

Example 1:
Height and *weight* of a growing child; the relationship is positive, since the child grows up, his height increases together with his weight.

- 2. In a **negative relationship**, one variable increases, the other variable decreases, and vice versa.

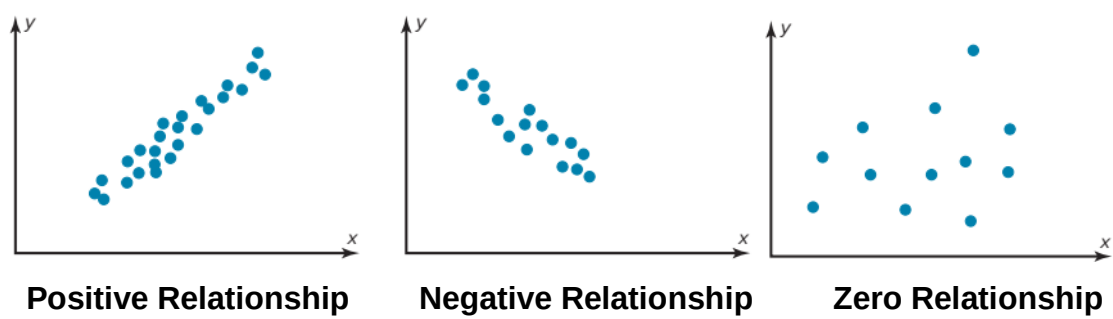
Example 2:
If you measure the strength of people over 60 years of age, you will find that as *age* increases, *strength* generally decreases.

- 3. While **Zero relationship**, an increase in one variable is not accompanied by either an increase or decrease in another.

Example 3:
IQ and *weight* of the people. Are heavy persons generally more intelligent? Or are lighter persons generally less intelligent?

The independent and dependent variables can be plotted on a graph called *scatter plot*. The independent variable x is plotted on the horizontal axis, and the dependent variable y is plotted on the vertical axis.

A **scatter plot** is a graph of the ordered pairs (x, y) of numbers consisting of the independent variable x and dependent variable y .
Here are the scatter plots of relationship:



Procedure in drawing a scatter plot:

- 1. Draw and label the x and y axes.
- 2. Plot each point on the graph.
- 3. Determine the type of relationship (if any) that exists for the variables.

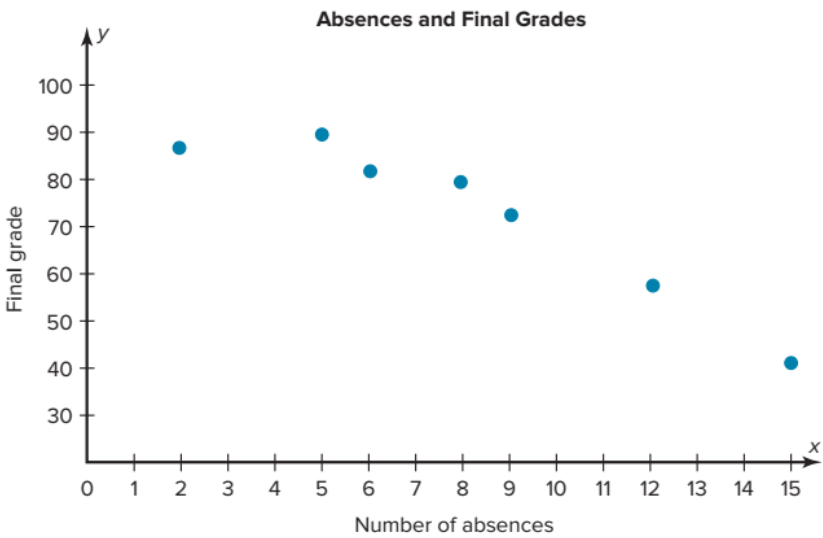
Example 4:

Construct a scatter plot for the data obtained in a study on the number of absences and final grades of seven randomly selected students from a statistics class. The results are shown below.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

Step 1: Draw and label the x and y axes.

Step 2: Plot each point on the graph, as follows.



Step 3: Determine the type of relationship (if any) that exists.

The graph looks as if a negative linear relationship exists between the number of student absences and final grade of the students.

4.3 PEARSON CORRELATION r

Generally, we have four types of correlations:

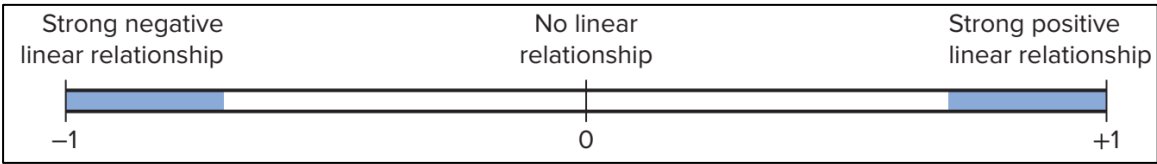
- Pearson Correlation,
- Kendall Rank Correlation,
- Spearman Correlation, and
- Point-Biserial Correlation.

Of these types of correlation, the Pearson Correlation and Spearman Correlation are the most widely used and will be discussed here.

The Pearson Correlation or also known as Pearson Product-Moment Correlation is a correlation statistic used to measure the degree of the relationship between linearly related variables. It was developed by Karl Pearson and it is used to measure the degree of linear relationship between the two variables. It is the covariance of the two variables divided by the product of their standard deviations.

The symbol for Pearson's correlation is " ρ " when it is measured in the population and " r " when it is measured in a sample. Since we will be dealing with samples, we will use " r " to represent Pearson's correlation. Pearson's r can range from -1 to 1 and may be either positive, negative or have no linear relationship between variables.

The data may be related in some other nonlinear way.

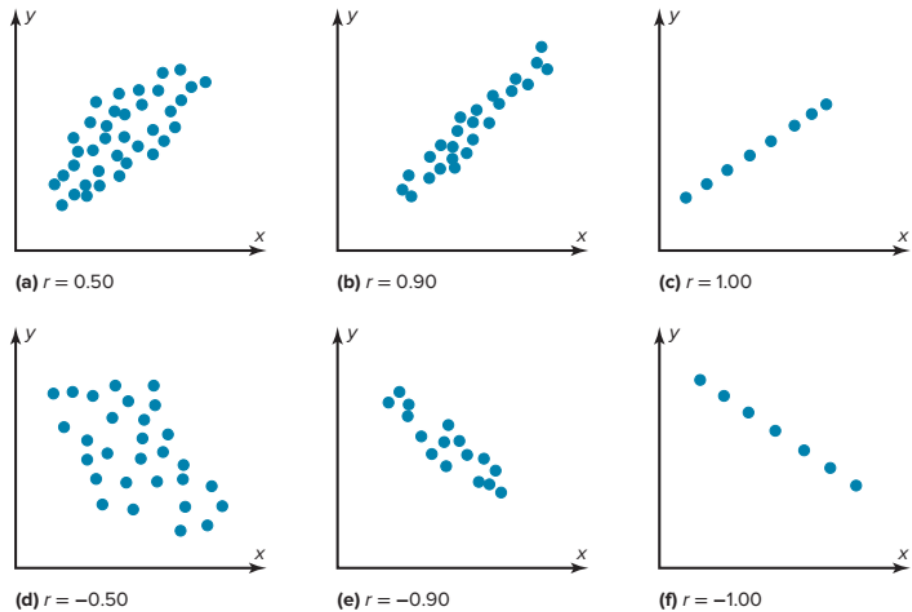


Ranges of Values for the Correlation Coefficient

The interpretation of r is shown in the table below.

Value of r	Interpretation
± 1 to ± 0.19	Negligible correlation
± 0.20 to ± 0.39	Low correlation
± 0.40 to ± 0.69	Moderate correlation
± 0.70 to ± 0.89	High correlation
± 0.90 to ± 0.99	Very high correlation

These are the relationship between the correlation coefficient and the scatter plot.



Properties of Linear Correlation Coefficient

- a. The correlation coefficient is a unitless measure.
- b. The value of r will always be between -1 and $+1$ inclusively. That is $-1 \leq r \leq 1$.
- c. If the values of x and y are interchanged, the value of r will be unchanged.
- d. If the values of x and/or y are converted to a different scale, the value of r will be unchanged.
- e. The value of r is sensitive to outliers and can change dramatically if they are present in the data.

Assumptions for the Correlation Coefficient

- a. The sample is random sample.
- b. The data pairs fall approximately on a straight line and are measured at the interval or ratio level.
- c. The variables have a bivariate normal distribution. (This means that given any specific value of x , the y values are normally distributed: and given any specific value of y , the x values are normally distributed.)

The formula for calculating Linear Coefficient Correlation r

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where n is the number of data pairs.

Example 5:

Compute the value of the correlation coefficient for the data obtained in the study of the number of absences and the final grade of the seven students in the statistics class given below.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

Solution:

Step 1: Make a table.

Student	Number of absences x	Final grade y (%)	xy	x^2	y^2
A	6	82			
B	2	86			
C	15	43			
D	9	74			
E	12	58			
F	5	90			
G	8	78			
	$\Sigma x =$	$\Sigma y =$	$\Sigma xy =$	$\Sigma x^2 =$	$\Sigma y^2 =$

Step 2: Find the values of xy, x^2, y^2 ; place these values in the corresponding columns of the table.

Student	Number of absences x	Final grade y (%)	xy	x^2	y^2
A	6	82	492	36	6,724
B	2	86	172	4	7,396
C	15	43	645	225	1,849
D	9	74	666	81	5,476
E	12	58	696	144	3,364
F	5	90	450	25	8,100
G	8	78	624	64	6,084
	$\Sigma x =$	$\Sigma y =$	$\Sigma xy =$	$\Sigma x^2 =$	$\Sigma y^2 =$

Step 3: Add the values in each column of x, y, xy, x^2, y^2 and record the results.

Student	Number of absences x	Final grade y (%)	xy	x^2	y^2
A	6	82	492	36	6,724
B	2	86	172	4	7,396
C	15	43	645	225	1,849
D	9	74	666	81	5,476
E	12	58	696	144	3,364
F	5	90	450	25	8,100
G	8	78	624	64	6,084
	$\Sigma x = 57$	$\Sigma y = 511$	$\Sigma xy = 3745$	$\Sigma x^2 = 579$	$\Sigma y^2 = 38,993$

Step 4: Substitute in the formula and solve for r .

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$
$$= \frac{(7)(3745) - (57)(511)}{\sqrt{[(7)(579) - (57)^2][(7)(38,993) - (511)^2]}} = -0.944$$

Step 5: Make an interpretation.

The value of r suggests a strong negative relationship between a student's final grade and the number of absences a student has. That is, the more absences a student has incurred, the more likely he would get a lower grade.

4.4 REGRESSION ANALYSIS

Regression analysis is a widely used technique which is useful for many applications. We introduce the technique here and expand on its uses in subsequent modules.

Simple Linear Regression

Simple linear regression is a technique that is appropriate to understand the association between one **independent (or predictor) variable** and one continuous **dependent (or outcome) variable**. For example, suppose we want to assess the association between study of the number of absences and the final grade of the seven students in the statistics class, where *final grade* is the dependent variable, and *number of absences* is the independent variable. In regression analysis,

- the *dependent variable* is denoted as y ; and
- the *independent variable* is denoted as x .

So, in this case y = final grade and x = number of absences.

When the two variables are strongly or highly correlated, then it is possible to predict or estimate the value of the dependent variable from the knowledge of the independent variable. This means that the measure of relationship or correlation coefficient between these two variables is ± 0.40 or higher. The prediction of the dependent variable denoted by y' is obtained using the formula:

Formulas for the simple linear regression $y' = a + bx$

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

where a is the y' intercept and b is the slope of the line.

Note: Round the values of a and b to three decimal places.

Example 6:

The number of absences incurred by students and their corresponding final grades are shown below. Find the following:

- A. Correlation Coefficient
- B. Regression Equation

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

- C. Based on the regression equation from B, predict the final grades of three students given the number of absences they have incurred.

STUDENT NO.	NUMBER OF ABSENCES (x)	FINAL GRADE (y)
H	3	?
I	7	?
J	10	?

Solution for A:

The Correlation coefficient of number of absences and final grade is $r = -0.944$ (see the solution in **Example 5.7**). The value of r suggests a strong negative relationship between a student's final grade and the number of absences a student has. Hence, *regression analysis is possible*.

Solution for B:

The values needed for the equation a and b are $n = 7$, $\Sigma x = 57$, $\Sigma y = 511$, $\Sigma xy = 3745$, and $\Sigma x^2 = 579$ as in **Example 5.7-step 3**. Substituting in the **formulas 5.2**, you get

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2} = \frac{(511)(579) - (57)(3745)}{(7)(579) - (57)^2} = 102.493$$

$$b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2} = \frac{(7)(3745) - (57)(511)}{(7)(579) - (57)^2} = -3.622$$

Now, using the equation of the simple linear regression $y' = a + bx$ from **formula**.
Hence, since $a = 102.493$ and $b = -3.622$ then the regression equation is $y' = 102.493 - 3.622x$.

Solution for C.

The predicted Final Grade of students given their number of absences shall be computed using the regression equation stated above, such as,

$$y' = 102.493 - 3.622x.$$

STUDENT NO.	NUMBER OF ABSENCES (<i>x</i>)	FINAL GRADE (<i>y</i>)
H	3	$y' = 102.493 - 3.622 (3) = 91.627$
I	7	$y' = 102.493 - 3.622 (7) = 77.139$
J	10	$y' = 102.493 - 3.622 (10) = 66.273$

Learning Activity:

- Draw the scatter plot for the variables given.
 - The number of murders and robberies per 100,000 populations for a random selection of states is shown.

Murders	2.4	2.7	5.6	2.6	2.1	3.3	6.6	5.7
Robberies	25.3	14.3	151.6	91.1	80	49	173	95.8

- The average gasoline price per liter (in cities) and the cost of liter of oil are shown for a random selection of weeks in 2020.

Oil (₱)	85	86	84	88	83	83	80	89
Gasoline (₱)	40	42	40	50	39	38	35	52

- School Administrators wondered whether class size and grade achievement (in percent) were related. A random sample of classes revealed the following data.

No. of Students	15	10	8	20	18	6
Average Grade (%)	85	90	82	80	84	92

2. Identify the relationship between the given two variables in items 1a, 1b, and 1c. You can refer to your scatter plot.
3. Give one example of variable in each type of relationships.
4. A researcher computed the correlation between the following variables. Interpret the results.
 - a. The I.Q. of brothers and sisters is 0.72
 - b. The I.Q. of husbands and wives is 0.34
 - c. The I.Q. of fathers and sons is 0.81

Learning Evaluation:

Answer the following:

1. The following table shows the amount of converted sugar in a chemical process at different temperatures.

Temperature, x	Converted Sugar, y
1.2	8.2
1.4	8.5
1.6	8.4
1.8	9.3
2.0	8.9
2.2	10.5
2.4	9.3

- a. Compute for the correlation coefficient.
 - b. Determine the linear regression line.
 - c. Estimate the mean amount of sugar produced when the temperature recorded is 1.7.
2. The grades of 9 senior high school students in a statistics course on a midterm period (x) and on the final examination (y) are as follows:

x	80	94	96	98	67	77	60	71	72
y	85	100	100	68	82	66	78	54	68

- a. Compute for the correlation coefficient.
 - b. Compute the regression equation.

References:

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